Only Means Only: what a harmonization benchmark score averages over

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Abstract

Image harmonization methods adjust the colour and lighting of an object pasted into a photograph so that it looks native to its new scene, and they are ranked by average error on one benchmark whose test images are real photographs with one object recoloured. One photograph can yield several such test images, and the benchmark promises that no photograph appears in both the training and the test split. We find that 1,399 photographs do, and the composites built on them are a quarter of the test set. Duplicate checks over file names or bytes miss this, because every composite is a different file. On that quarter the three methods we score end to end read two to five times lower error than on the rest, but most of the gap is composition rather than the sharing: the shared composites come almost entirely from one of the benchmark's four source collections, and inside that collection the gap falls to about one and a half times. An official score is therefore an average over two populations, mixed in a proportion the benchmark does not report. We release a test list of 5,581 composites whose construction identities do not appear in training, with four confirmed near-duplicate pairs disclosed rather than removed, adoptable without retraining at the price of one rescore per checkpoint. We leave open whether the sharing itself changes what a model learns.

1 Introduction

Image harmonization has advanced largely by scoring methods on one benchmark. iHarmony4 [1] takes real photographs, alters the colour and lighting of one foreground object in each so that it looks as if it came from another scene, and releases a fixed train and test split of those altered copies. Methods are ranked by error against the unaltered photograph. The benchmark has four sub-datasets, and HCOCO, built from COCO, holds most of both released lists. A score on such a benchmark is a statement about an evaluation population as much as about a method: it depends on which composites the test split contains and on where those composites came from.

For a composited benchmark that dependence is not incidental. One real photograph can back several composites, and the released split decides which of them are training and which are test. Figure 1 shows the case in two composites built on one photograph, recoloured differently, one on each side of the split. When a test composite is built on a photograph the training split also uses, the model has already seen the pixels it is asked to reproduce, including the foreground

object whose unaltered appearance is the answer, and the reported score blends such composites with genuinely unseen ones. The standard protocol does not check for this. Its hygiene measures, such as de-duplicating file names or file bytes, operate on composites, and two composites built from one photograph are different files. The unit that has to be disjoint is the real photograph a composite is built from, because that photograph is also the answer, and that unit is rarely audited before scores are compared.

We audit iHarmony4 at that construction unit. The photograph behind a composite is recovered from its file name: the name carries the target photograph's identifier first, then the reference image used to recolour the object, then a variant index. The official test and train lists have 1,399 target photographs in common, and those photographs account for 1,822 of the 7,404 composites in the test list, 24.6 percent, each built on a photograph that also appears in training. One further composite is removed because its target photograph is pixel-identical to a training target under a different name, so 1,823 composites are removed in total. Removing them leaves a retained test subset of 5,581 composites whose target photographs do not appear in the training list, four confirmed near-duplicate pairs aside, and the training list itself is left untouched, so fixed published checkpoints can be rescored without retraining. We then run controls to determine what the sharing does and does not do to the comparisons the benchmark is used for. Adopting the retained list is a substitution of the evaluation list: it costs one rescore per checkpoint, and a score on it is not directly comparable with a score on the official list.

What the sharing does to reported numbers is the harder question, and the answer is more structured than a single effect size. Pooled over sub-datasets, the excluded composites read two to five times lower mean squared error than the rest for the three methods scored end to end, which looks like a large advantage for a model that has seen those photographs. Inside HCOCO, where most excluded composites sit, the same ratios fall to 1.47 to 1.61, and the DoveNet-Harmonizer gap narrows there instead of widening (post-hoc). The ratios are of whole-image mean squared error (MSE), and the foreground-region version of the same metric, computed over the pasted object's region and written fMSE here, is the one harmonization papers report. A foreground-ratio-stratified control in MSE is negative for Harmonizer and for our own editor and is not detected for DoveNet, and inside HCOCO, where the mixture cannot differ, none of the three is detected at floors of 5.9 to 15.7 MSE. One published checkpoint, DiffHarmony [2], moves the other way on MSE and scores worse on the excluded composites. The pooled difference is therefore largely arithmetic over two differently composed populations, 99 versus 44 percent HCOCO, rather than a measurement of what the sharing does to a method.

What the sharing does to what a model learns is not measured here. A pre-registered audit of four published checkpoints ran below its registered five-checkpoint floor, with the retained-versus-excluded interaction significant for all six method pairs after Holm correction on the pooled lists. The pre-registered exposure retrain departed from its registered execution, and both of its intervals include zero (Section 4.4). The evaluation itself can be corrected today: the retained list is ready to adopt.

Contributions. (1) A construction-unit audit of the released iHarmony4 lists, with deterministic test-side exclusions and a perceptual residue check whose blind spots we state (Sections 3 and 3.1). **(2)** A construction-identity-disjoint retained test subset and its excluded counterpart, kept beside the official list so that fixed published checkpoints are rescored without retraining, together with the five evaluation-hygiene rules the audit motivates (Sections 5 and 5.2). **(3)** A controlled score-sensitivity analysis, reported as observational: fixed-checkpoint population contrasts, foreground-ratio-stratified controls, a below-floor four-checkpoint audit, and an exposure retrain whose arms departed from the registered execution and which therefore establishes neither a difference nor practical equivalence, each with the composition and execution limits

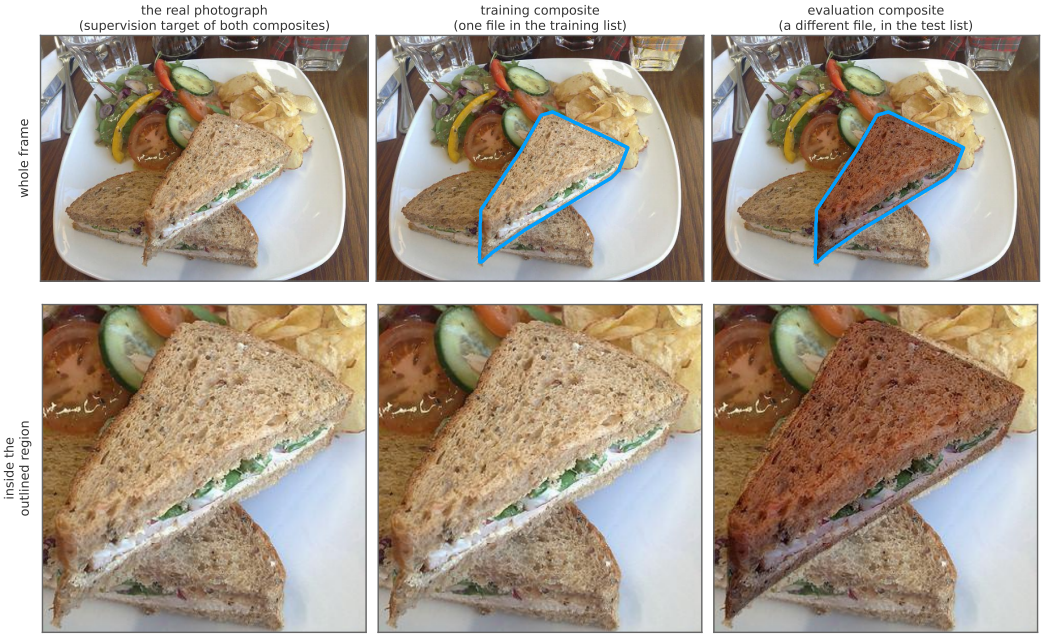


Figure 1: One photograph, two composites. Both are built from the photograph on the left, the answer the model must reproduce: the object outlined in blue has been recoloured differently in each. The two files share no name and no bytes.

stated beside the result (Section 4). Tables and derivations are in the supplement. Section 9 states what regenerates.

2 Evaluation populations

Removing the composites built on a shared photograph from a test list changes two things at once, and a reported score alone cannot say which of them produced the change. The first is the mixture. The composites built on a shared photograph are overwhelmingly HCOCO, so the retained list is not the same benchmark with fewer errors: it is a differently weighted benchmark, and a method that is stronger on HCOCO will move for that reason alone. The second is the sharing itself. On the retained list no composite is built on a photograph the model has seen, so a method that has effectively memorized the photograph it must reproduce loses that advantage. Both would lower the score of such a method, and they are not separable from the score. The mixture can be held fixed inside one sub-dataset, but the evidence against the sharing is the absence of a detected difference, and an absence of detection is not an absence of difference.

A third possibility sits between them, and it is easy to miss because it appears as a difference of convention rather than of population. Two evaluations of one checkpoint on one list can report different numbers when they disagree about the metric definition, the peak used in PSNR, the working resolution, the region of integration, the compositing rule or the batch-normalization mode. None of those is a property of the data, and each can be as large as the effect we set out to measure.

We separate the three by construction rather than by argument. For the mixture we compare inside a single sub-dataset, where the mixture cannot differ, and we stratify by the foreground ratio at which the metric is evaluated, the fraction of the frame covered by the recoloured object, (Sections 4.1 to 4.3). The convention question is not a rival explanation: it is a property of our measurement, and it is answered by a reproduction check that opens the results. For the sharing

itself no fixed-checkpoint analysis can substitute for an intervention, because every checkpoint we score was trained with the sharing present. We ran the intervention and report it with its execution departures (Section 4.4).

3 Shared photographs in the official lists

One real photograph can back several composites. The original dataset description promises that a target image is not shared across splits: "We ensure that the same target image does not appear in the training set and test set simultaneously, to avoid that the trained model simply memorize the target image" [1]. The released lists contain composites that stand on the same real photograph across the two splits: the shared element is the construction identifier carried in the filename, so what crosses the split is the real photograph the composite is built from, which is also the answer the model must reproduce, not the composite file itself. A duplicate check over file names or file bytes does not see it, because the composites are different files. Overlap between evaluation and training data is a known problem in classification, where duplicates and near-duplicates inflate measured accuracy [3, 4], and it is one of the leakage classes that destabilise published results [5]. Supplement S10 gives a worked example with its positions in the released lists. The overlap is a property of the two lists, and the construction explains how it arises: a composite is one target photograph paired with one reference image, the 42,828 HCOCO composites are built on 28,938 target photographs, so a photograph used as a target more than once carries more than one composite, and the released lists place some of those composites on each side of the boundary. The promise quoted above names the target image, and this is the case it rules out. The counts below come from the released training and test lists as the dataset repository distributes them. Supplement S10 records the two list files and their content hashes. Our matcher derives the target-photograph identity of every composite from these lists and refuses to proceed if the layout it is handed still shares identities across splits. Section 9 states which counts this reproduces and which it does not.

We call a composite shared when the photograph it is built from appears in both lists. 1,399 target photographs are shared, and the composites built from them account for 1,822 of the 7,404 composites in the official test list, so 24.6 percent of the test set is built on a photograph that also appears in training. Removing them leaves a retained test subset of 5,581 composites with no target photograph in common with the training list, and the training list is left untouched, so published checkpoints stay eligible for rescoring without retraining. One further composite is removed only by an exact decoded-pixel duplicate check, which finds exact decoded-pixel copies only. The per-sub-dataset decomposition, the validation split carved from the training list under the same identity rule, and the staged counts are in Supplement Table S12. The removal is deterministic list arithmetic, and the excluded composites are the set difference between the official and retained test lists.

Other composited benchmarks. A post-hoc construction scan of public split lists (name level, CPU only, not a pre-registered audit) did not find the iHarmony4 pattern in the two other benchmarks it could scan. ccHarmony [6] (3,080 train and 1,180 test composites over 250 and 100 target photographs) shares 0 identities between train and test and 0 with the iHarmony4 official train identities, and no real or mask file crosses its split. DESOBA [7] (839 train and 160 test source images) shares 1 scene stem, 1 of 160 test images (0.6 percent), read as near-duplicate scene content (shifted viewpoint or crop, grayscale correlation 0.35, one pair inspected by eye). iHarmony8K, DESOBAv2, SRD, USR and WSID were blocked (no source found, or archive too

large to fetch). ISTD+ and HYouTube were not scanned (Supplement Table S11). The scan does not test near-duplicates under different names, pretraining overlap or scores.

3.1 Limits of identity matching

The matcher keys on identity strings, so two frames of one scene, two crops of one photograph, or a re-encoded duplicate under a different filename would not be matched. A 64-bit dHash over every unique target photograph (pool sizes and the full radius table in Supplement Table S7), test queried against train at Hamming radii $\{0, 2, 4, 6, 8\}$ with every Hamming-0 pair confirmed by a 64×64 grayscale correlation, returns at Hamming 0 four cross-split pairs beyond the registered exclusions (1 HCOCO, 1 HAdobe5k, 2 HFlickr): three are identical content ($r \geq 0.9999$, recompression under a different filename) and one is same-scene ($r = 0.9509$). These are 4 of 2,917 target photographs in the retained test list, a confirmed floor of 0.14 percent and not the size of the residue. They remain in the retained list. Wider radii return candidates (5 at ≤ 2 bits, 8 at ≤ 4, 19 at ≤ 6, 78 at ≤ 8) that were not adjudicated. The retained list is therefore disjoint by path identity only and is not a clean split. A whole-matrix screen extends that check beyond the hash radii: every one of the 7,404 test composites is compared with every one of the 65,742 training and validation composites at 32×32 grayscale correlation, and cross-split pairs above 0.99 whose identity strings differ are reported. It returns six candidates. One is byte-identical, and its composite is one of the composites removed by the decoded-pixel check. The other five sit in the retained list at correlation 0.99997 or above, so the retained list carries them (Supplement Section S4).

4 Effect on reported scores

Before reading any difference as population, we fixed the conventions, because two evaluations of one checkpoint on one list can disagree without any population changing. Our scoring reproduces the numbers its authors published: to within 0.13 MSE for DoveNet once its upstream batch-normalization setting is used, and to within 1.2 percent fMSE for Harmonizer under its authors' own evaluation code, and every table in this paper uses one protocol for every method (Supplement Sections S18 and S17).

How large is the difference between the two populations in the reported metric? How much of that difference is composition rather than the sharing? Does it reproduce across methods and inside a single sub-dataset, and does it survive a control that matches the foreground ratio at which the metric is evaluated? Does an audit or an intervention establish it? And does an external benchmark confirm the retained list's advantage?

4.1 Pooled error on the shared subset

Method (MSE ↓)	ret. 5,581	excl. 1,823	offic. 7,404	Δ excl.−ret.
DoveNet	140.18	69.40	122.75	−70.78
Harmonizer	133.42	26.43	107.08	−106.99
Editor (ours)	149.45	48.82	124.68[†]	−100.63

Table 1: Whole-image MSE on the retained, the excluded and the official list; the shared quarter reads 2 to 5 times lower for every method. [†]Weighted mix of the two lists rather than a separate scoring (Supplement Table S19 in Section S14).

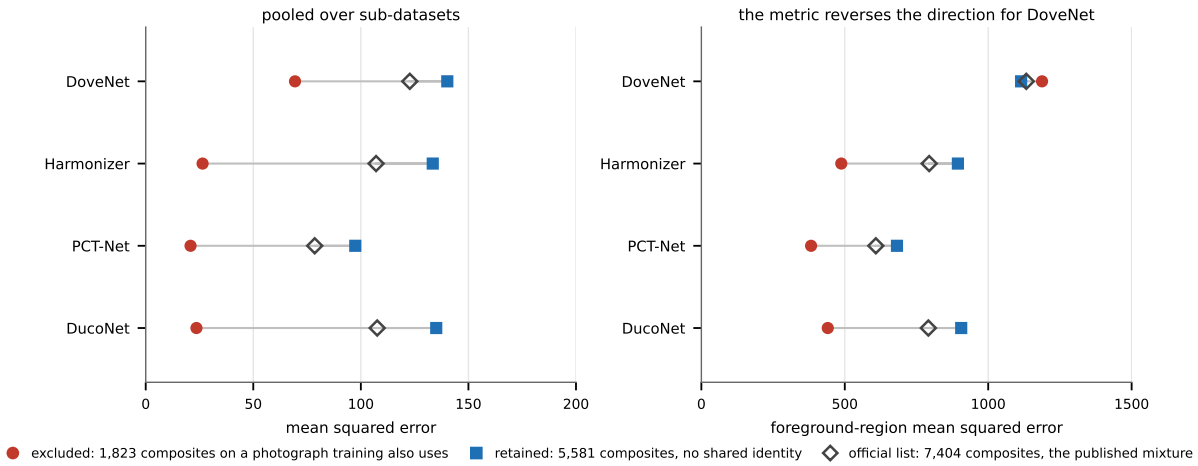


Figure 2: An official score is a mixture of two populations. Each checkpoint is plotted on the excluded composites (circle), the retained composites (square) and the official list (diamond), in whole-image MSE at the left and foreground-region MSE at the right.

Every method's official-list number mixes roughly 75 percent retained and 25 percent shared composites. Pooled, the shared quarter scores 2.02, 5.05 and 3.06 times lower in MSE (DoveNet, Harmonizer, ours). The excluded list is 99 percent HCOCO (1,807 of 1,823), the retained list 44 percent (2,476 of 5,581). Inside HCOCO the factors are 1.47, 1.61 and 1.58 with intervals above 1, not foreground-matched (post-hoc, Supplement Table S19). Figure 2 draws that arithmetic for the four checkpoints whose scores are recorded on all three lists: the official-list number sits between the two populations it averages, and on the foreground-region metric the direction turns over for DoveNet.

The direction is not shared by every checkpoint we scored. Three further published checkpoints, scored for the audit of Section 4.4, give retained-to-excluded MSE ratios of 4.68 for PCT-Net [8] and 5.73 for DucoNet [9] (1.56 for both inside HCOCO), and 0.83 for DiffHarmony (0.90 inside HCOCO): DiffHarmony scores worse on the shared composites in both populations (Supplement Table S19). PCT-Net and DucoNet were subsequently scored on the official list as well, so all three populations are now measured for them: their official-list MSE is 78.56 and 107.60, between their retained values (97.41 and 135.04) and their excluded values (20.83 and 23.58), which is the mixture arithmetic this section describes, on two further checkpoints (Supplement Table S5). DiffHarmony is outside the frozen four-checkpoint audit family and is reported descriptively. It is built on a pretrained Stable Diffusion inpainting model whose pretraining data a list-based audit cannot examine.

The fMSE column moves differently and we state it: DoveNet is +73.0 fMSE *worse* on the excluded composites, Harmonizer −406.5 better, ours −233.0 better. That is consistent with the compositing-metric asymmetry of Section 6: the excluded composites carry smaller foregrounds on average, and fMSE normalizes by exactly the region that is smaller.

Source of the pooled difference. On the retained list the DoveNet-minus-Harmonizer MSE gap is 6.76. On the official list the same two checkpoints, scored by the same driver, read 15.67, 2.3 times larger. This pooled ratio is not a measure of exposure: its interval is uninformative ($[-13.42, +15.16]$). Inside HCOCO the official-to-retained ratio of that gap is 0.88 $[0.84, 0.94]$ (51.77 against 58.73), and the excluded-list gap is 42.25, lower than both, so the shared composites narrow it. Removing 1,807 random HCOCO composites (plus the 16 other excluded ones) gives

pooled ratios with median 4.25 (3.05 to 7.57), above the observed 2.3 (post-hoc, a reference rather than a matched placebo, Supplement Table S20 in Section S14). Between these two baselines Harmonizer leads on all three lists and no ranking flip occurs.

Foreground-ratio stratification and its floor. For each test composite, the dose k counts the training composites built on the same target photograph, so $k = 0$ on the retained list by construction and $k \geq 1$ on the excluded one (the dose distribution over the 7,404 composites is in Supplement Section S1).[1] Composites are binned on foreground ratio (eight pooled quantile bins), and the contrast is the within-bin $k \geq 1$ versus $k = 0$ step. The estimator uses list membership only: doses above 1 are not used, and sub-dataset is not matched (Section 4.2).

For Harmonizer the unmatched pooled step ($k \geq 1$ minus $k = 0$) is -106.98 MSE, and the within-bin step is negative in every bin (8/8). This is not the -106.99 of Table 1, which splits the composites 1,823 / 5,581 instead of 1,822 / 5,582. For DoveNet the pooled step is -70.90, but within bins every step reverses sign (8/8 positive), a direct demonstration that the pooled contrast was population composition (a negative step means shared composites score better, per-bin ranges in Supplement Table S1). The same reading applies to the pooled matched steps (Section 4.2). Within-dose slopes are unstable in sign for both methods at these sample sizes and no trend is claimed from them.

4.2 Pooled and within-HCOCO steps

All three methods go through the identical statistic: dose k as above, the bin-averaged $k \geq 1$ minus $k = 0$ step within eight foreground-ratio bins, and a cluster bootstrap over target photographs, with the bin-averaged step, its interval and its 2.8σ detection floor reported once. The pooled rows are the analysis as first run (4,317 identities, 400 draws). The HCOCO-only rows re-run the same estimator on the same records (post-hoc, 3,417 identities, 2,000 draws, Supplement Table S18 in Section S14).

Method	Rows	Step (MSE ↓)	95% CI	Detection limit
DoveNet	pooled	+16.36	[−5.0, +40.8]	33.8
	HCOCO only	+2.08	[−8.5, +13.4]	15.7
Harmonizer	pooled	−62.31	[−73.6, −49.4]	17.5
	HCOCO only	−2.86	[−7.0, +1.3]	5.9
Editor (ours)	pooled	−30.40	[−47.9, −9.5]	26.8
	HCOCO only	−2.32	[−10.0, +5.8]	11.0

Table 2: The matched exposure step under one estimator for all three methods, pooled and inside HCOCO. A negative step means the shared composites score better once foreground ratio is matched (Supplement Figure S1). Detection limit: 2.8 bootstrap standard errors.

Pooled, Harmonizer (-62.31 MSE) and our typed editor (-30.40) have intervals wholly below zero. DoveNet's step is positive and inside its 33.8 MDE. The pooled estimator matches foreground ratio but not sub-dataset: of the 1,822 exposed composites 1,806 are HCOCO and 16 HFlickr, against 812 unexposed HFlickr composites (HAdobe5k and Hday2night have none exposed), and the unexposed side (5,582) is 44 percent HCOCO (Supplement Table S21 in Section S14). Inside HCOCO, the only sub-dataset with enough exposed composites for a

[1] The $k = 0$ count is 5,582 rather than 5,581 because the one decoded-pixel-duplicate exclusion (its target is pixel-identical to a train target under a *different* filename) never occurs in the train identity list and so has $k = 0$. The remaining 1,822 exclusions are path-identity matches with $k \geq 1$.

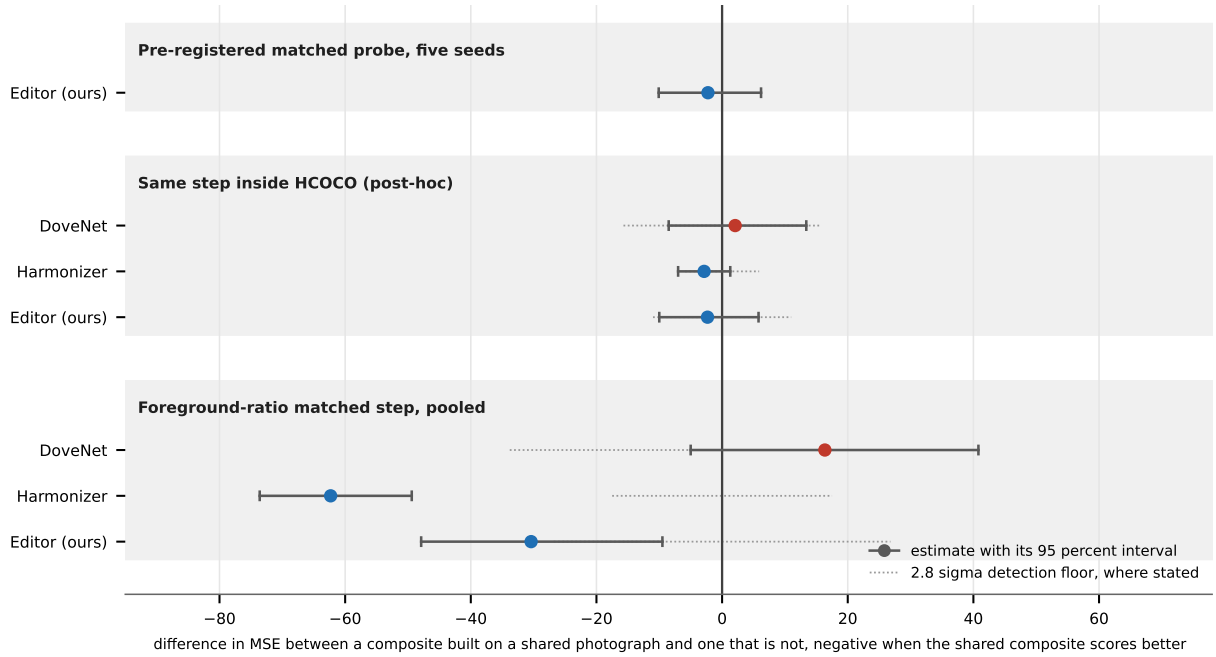


Figure 3: The shared-composite step in whole-image MSE, so the intervals are comparable. It is large and negative, pooled, for Harmonizer and for our editor. Inside HCOCO every interval includes zero, with the detection floors drawn at 5.9 to 15.7 MSE.

within-sub-dataset contrast, every interval includes zero, and Harmonizer's excludes any step below −7.0 against the pooled −62.31. The pooled-minus-HCOCO difference is 95 percent of Harmonizer's pooled step and 92 percent of ours, with intervals excluding zero (DoveNet's pooled step was not detected under this estimator). Figure 3 collects the MSE contrasts for the three methods scored end to end, on the shared photographs on one axis, drawn against the detection floor of the design that produced it.

The same estimator on the three further checkpoints (post-hoc, Supplement Table S18) gives pooled and HCOCO-only MSE steps of −42.49 and −1.40 for PCT-Net and −69.35 and −1.72 for DucoNet, the pattern of the table.

One direction does not hold. DoveNet's pooled matched step is positive (+16.36, inside its 33.8 floor) while Harmonizer's and ours are negative and clear their floors (−62.31 and −30.40). Inside HCOCO the three steps are +2.08, −2.86 and −2.32, all including zero. DiffHarmony goes further and is detected inside HCOCO: +50.23 [+41.0, +59.3] pooled and +26.76 [+19.5, +34.3] there, so shared composites score *worse* for it. The reversal tracks the metric. On fMSE, DoveNet is +73.0 worse on the excluded composites while Harmonizer is −406.5 better and ours −233.0 better (Section 4.1). The same population moves an error metric in opposite directions for different checkpoints, so the pooled step is a property of the checkpoint, its training objective and the region each metric integrates, not a population-wide exposure signal.

In fMSE (not in the table), the post-hoc within-HCOCO residual is not used as evidence for the main claim: under the stated multiplicity families, only Harmonizer and PCT-Net survive the narrower family among the frozen-family methods, and neither survives the broader family (Supplement Table S23). The residual may reflect foreground size. The registered audit and the retrain that follows it (Section 4.4) use separate estimands and populations.

4.3 Our own editor under the same protocol

The third method is the one whose pipeline we control completely, so the model can be held byte-identical while only the population changes: the five per-seed typed checkpoints of Section 6, bound by SHA-256, are scored once over each population, inference only, with no retraining, gate, threshold or selection, and the excluded list had not been scored by this project before that pass. The control was pre-registered with its rule written down before the numbers: if the foreground-matched gap collapses toward zero, the raw contrast is population difficulty and no identity claim is licensed.

The raw contrast is large and in the same direction for every seed: the gap in mean absolute error inside the object's region, the editor's own quality gate, is positive in 5/5 seeds (+1.954 to +2.938). At $n = 5$ the smallest attainable two-sided sign-test value is $p = 0.0625$. MSE is 3.06× and fMSE 1.27× lower on the excluded quarter. Within HCOCO, over eight foreground-ratio quantile bins and a 2,000-draw cluster bootstrap over 3,417 target identities, the matched excluded-minus-retained interval includes zero on all three metrics (MSE -2.25 $[-10.09, +6.20]$, fMSE -42.77 $[-97.12, +15.56]$, PSNR $+0.04$ dB $[-0.22, +0.30]$). The excluded composites are also easier before any model touches them (do-nothing reference and the full table: Supplement Table S3). The pre-committed rule applies: for our own checkpoints, the target photograph is not separable from difficulty at this sample size. This population is HCOCO only, and on it the statistic of Section 4.2 reads the same for all three methods (MSE steps +2.08, -2.86 and -2.32, intervals including zero, and ours agrees with this probe's -2.25).

The arithmetic claim stands: an official-list score is a weighted average over two populations that these methods score very differently, so official-list and retained-list numbers are not interchangeable.

4.4 The registered audit and the intervention

In the pre-registered four-method fMSE audit (below its registered five-checkpoint roster floor), the retained-versus-excluded interaction gate passed after target-photograph identity bootstrap, foreground matching, and Holm correction. All six fMSE pair interactions in the frozen DoveNet, Harmonizer, PCT-Net and DucoNet family were significant after Holm correction, and their intervals excluded zero under pooled foreground matching. The audit compares the retained 5,581 composites with the excluded 1,823 (2,000 target-photograph identity bootstrap replicates, seed 0, $\alpha = 0.05$, eight pooled foreground-ratio bins).

The registered audit reproduced the previously measured pair contrast: the DoveNet-minus-Harmonizer fMSE gap widened by +479.50 on excluded versus retained composites (95% CI $[403.73, 553.03]$ from the seed-0 2,000-draw run, +504.60 $[423.65, 583.84]$ foreground-matched; Supplement Table S20 gives the same point estimate under its per-stratum reseeded run, $[+403.61, +557.22]$), against the pre-registered +479 point prediction. That prediction was computed from shifts already measured on the same two checkpoints (+72.96 and -406.54 fMSE), so the point value repeats an earlier measurement. What the audit adds is the clustered interval, the Holm family and the PCT-Net and DucoNet checkpoints. Post-hoc, inside HCOCO alone, the interaction is -10.68 $[-77.97, +56.26]$, so the pooled interaction is not detected within the one sub-dataset where it can be estimated (Supplement Table S20). Pairwise gaps depend on the evaluation population: DoveNet-minus-Harmonizer is +699.88 on excluded and +220.38 on retained composites. Harmonizer-minus-DucoNet changes from +47.16 to -11.53, a descriptive point-estimate reversal.

Execution of the intervention. Both evaluation difference-in-differences (DiD) intervals span zero and their design departed from the registered execution, so neither the difference nor practical equivalence is established. DoveNet's exposure DiD was −8.37 fMSE (95% CI [−41.61, +23.90], MDE 47.09 fMSE under the 2.8×SE convention) and Harmonizer's +8.23 ([−37.78, +59.64], MDE 65.22). Table 3 keeps the fMSE between-arm change and the training-check manipulation-check contrast separate from the audit and matched-step tables.

The registered design trains DoveNet and Harmonizer in two arms per method and three paired seeds, with 63,226 training composites per arm including a matched 1,962-composite arm set (registered budgets, epochs and steps in Supplement Section S12). The two arms are the full-list arm, which keeps the 1,962 exposed composites in training, and the reduced-list arm, which replaces them with a seeded, sub-dataset-matched sample of unexposed composites; both arm lists carry 63,226 composites: a common 61,264-composite unexposed base plus the arm's own 1,962 composites, drawn from the 65,742-row official training list. The estimand is exposed-minus-rest fMSE in the full-list arm minus the same contrast in the reduced-list arm, stratified by subdataset and averaged over seed pairs. On the test side, exposed comprises 1,874 composites and rest 5,530 retained composites after removing 51 exposed overlaps. Uncertainty uses 2,000 joint-bootstrap replicates over seed pairs and target identities within subdataset, seed 0. The fixed equivalence margin is (−25, +25) fMSE. Neither interval lies inside that margin. Three seed pairs give coarse seed uncertainty. MDE is 2.8 times bootstrap SE, an approximate detection limit.

Method	Estimand	fMSE contrast	95% joint-bootstrap CI	MDE
DoveNet	Change between arms in the shared-composite advantage	−8.37	[−41.61, +23.90]	47.09
Harmonizer	Change between arms in the shared-composite advantage	+8.23	[−37.78, +59.64]	65.22
DoveNet	Training check on the arms' own composites	−308.75	[−409.88, −234.13]	n/a
Harmonizer	Training check on the arms' own composites	−56.51	[−69.96, −41.08]	n/a

Table 3: The registered intervention and the training check, three seed pairs per method. The training-check contrasts compare the full-list and reduced-list arms on the frozen 1,962-composite manipulation population, and both exclude zero.

The training-check contrasts of Table 3, with all three seed pairs contributing for each method, show that the two arms' checkpoints differ on composites that were in one arm's training set and not the other's. They support that the training manipulation affected performance on its specified population. The training check was scored after the pre-registration tag, a disclosed deviation, and does not alter the registered verdict calculation. Provenance of the intervention and training-check values is detailed in Supplement Section S9.

Sensitivity reads. The frozen audit gate passes under fMSE, MSE and PSNR (fMSE binding), with six, four and five of six pairs Holm-significant and all six interaction directions in agreement after orienting PSNR to the error direction. The Harmonizer-DucoNet point-estimate reversal occurs under fMSE and MSE, and PSNR has none.

4.5 External comparison on ccHarmony

The same four checkpoints were evaluated zero-shot on the 1,180 ccHarmony test composites over 100 target photographs. At pairwise sign level, agreement with ccHarmony is 5 of 6 for the retained list and 6 of 6 for the official list (the 7,404-composite union of retained and excluded rows), so $\Delta_{\text{agreement}}$ is -1 of 6. Dropping either DucoNet or Harmonizer moves the leave-one-method-out (LOMO) sign to zero. The retained-versus-ccHarmony Kendall tau is 0.6667 $[0.6667, 1.0]$. The pre-registered prediction that retained-list rankings agree with zero-shot ccHarmony behavior at least as well as official-list rankings, is mixed at sign level and is not supported at the observed point estimates. The comparison is descriptive only: it had no gate or success threshold, it carries no power claim, and it does not show external validity of the retained list (quantities in Supplement Table S8, with signs on point gaps).

For a reader of published numbers, the pooled differences in this section are arithmetic over a mixture the benchmark does not disclose. The reported performance difference between two public methods is 2.3 times larger on the official list than on the retained list in point estimates. Inside HCOCO the same gap is smaller on the official list than on the retained list, so the shared composites narrow the better method's advantage there (Section 4.1). The DoveNet-Harmonizer gap is positive in point estimate in every population we scored; its interval excludes zero in all of them except the pooled retained list. The Harmonizer-minus-DucoNet gap does change sign in the descriptive point estimates (Section 4.4). A number on this benchmark is averaged over a population whose composition decides part of it, and a method that benefited from shared photographs would be rewarded for that rather than for harmonization.

5 The retained list

5.1 Adopting the retained list

The retained list is a replacement test list of 5,581 composites, kept beside the unmodified official list, so the 1,823 excluded composites are the set difference of the two and the training list is unchanged. In the project's development repository it comes with the identity matcher and its manifest, regression tests, and a re-derivation driver with a pin registry. The artifact repository shipped with this paper carries the matcher, the lists and the CPU checks, not the driver. The driver re-scores a public baseline on the official, retained or excluded list from pinned upstream commits and checkpoint digests, refuses to run against an unpinned or mismatched checkpoint, and checks the staged tree against both lists before scoring (a check that caught a real gap in our own staging).

Adopting the list means pointing the evaluation at the retained list instead of the official list and reporting the retained number beside the official one rather than in place of it. On the same pinned checkpoints scored by the same driver, retained MSE is worse for both methods (DoveNet 122.75 $\rightarrow$ 140.18, Harmonizer 107.08 $\rightarrow$ 133.42) and the pooled method-to-method MSE gap shrinks 2.3$\times$ in point estimates ($+15.67 \rightarrow +6.76$; the ratio interval $[-13.42, +15.16]$ includes 1 and the retained gap interval $[-5.20, +18.74]$ includes zero; Section 4.1). On fMSE the two methods move in opposite directions, the foreground-size difference of Section 4.1. The columns are not interchangeable and the difference is not a uniform correction (full table with fMSE, PSNR and changes: Supplement Table S13).

5.2 Evaluation-hygiene rules

The audit and a compositing-metric defect we found in our own trainers lead to five short rules. **H1. Match identities, not files.** File-level deduplication does not detect sharing of the real photograph a composite is built from. Match on the identity that generated the sample, fail closed, and audit the perceptual residue rather than assuming it is zero (Section 3.1). **H2. Correct the evaluation list, not the training list.** Where leaked training images cannot be removed, correcting the test side is the cheaper remedy [10]: it keeps every published checkpoint eligible for rescoring, while a correction that requires retraining is costlier to adopt. **H3. Score a composited editor after compositing.** Scoring the pre-composite tensor describes a different system with no locality guarantee. It moved one of our reported metrics by 8.2 dB in the wrong direction (Section 6). **H4. Disclose the compositing-metric asymmetry.** Whole-image MSE and PSNR mechanically reward a composited editor, since every unedited pixel is exactly right, while a full-image baseline pays for every pixel. A benchmark mixing the two should say so and lead harmonization-quality comparisons with fMSE, as we do, and on it our editor is not the lead row. **H5. Declare wear and score final numbers once, on a list-defined population,** after reporting how much of the test split the development run already touched and pre-committing the remedy.

6 The third method and its certificate

The results above need one method whose training data, compositing rule and evaluation convention we can inspect end to end, because the published baselines do not all expose them. We therefore train a small harmonization network of our own and freeze it as a third checkpoint. It is a control in this study rather than a harmonization-quality claim.

The control is useful here for one reason: its channel from outside the edited region to the pixels written inside it is bounded, and the bound can be computed rather than assumed. Written out in words, the network admits outside information through two paths. The first is a set of six global colour statistics of the whole frame, and the second is a pooled feature of the frame that we disable in the reported configuration. The output inside the scored region is a sum of a local path that reads only the trusted window, a physical term driven by those six statistics, and a semantic term driven by the pooled feature. For the architecture we ship, the worst-case change that any perturbation outside the trusted window can induce inside the scored region is bounded by a product of per-layer operator norms, and with the pooled path disabled that bound is exactly zero for the semantic term. The bound is proved for the stated architecture in Supplement Section S15, with an outline in Section S11, and it is checked in the repository three ways: closed-form re-derivation, regression tests at odd resolutions under outside-supported perturbations, and a negative control that can fail. Those checks are consistency checks, not a proof. The bound is loose by about a factor of thirteen, and a held-out projected-gradient search over 500 attacks never exceeded it (Supplement Section S6). Two properties should be kept apart: the composite returns the source pixel exactly where the effective write support is zero, which is a property of the write set and not of the bound, and the bound above is a property of the influence path, not of the write set. We call the control *typed* because its outside channels are named and separated, physical and semantic, each with its own bound.

Metric choice. Our editor writes inside a feathered region and composites elsewhere, so every unedited pixel is exactly right, and whole-image MSE and PSNR reward that convention mechanically while a baseline that emits a full image pays for every pixel. Only composited values computed after the scoring fix of 2026-08-21 appear in this paper. Before that fix our

own trainers reported the raw source-plus-residual on a 128-composite check, which described an editor with no such guarantee: MSE 572.62 and PSNR 21.83 dB against doing nothing at 325.72 and 30.01 dB, 8.2 dB worse, with mean background leakage of 15.685/255. After compositing, the same checkpoint reads MSE 267.87, PSNR 30.40 dB and fMSE 1550.55, and the mean absolute error inside the object's region, the pass/fail gate of the editor's own training check, moves only from 29.592 to 29.662, so no verdict changed. Every pre-fix value is invalid and is cited only here. Every table in this paper therefore states its metric, and rule H4 of Section 5.2 follows: the population arithmetic is reported in whole-image MSE, because the two populations are the object of study, and the comparisons of harmonization quality are reported in fMSE.

Placement of the third method. On the standard protocol, every method scored on the identical retained list, the editor is level with DoveNet on fMSE (1104.82 against 1114.71) and behind Harmonizer (894.34), and its error inside the object's region falls from the no-op source's 29.64 to 25.01 [24.45, 25.56]. That places a demonstration-scale network in the benchmark. We make no quality comparison with recent generative harmonizers [2, 11]. The placement table is Supplement Table S10. The paired ablation that measures what the certified channel is worth against a capacity-matched floor, and the price paid for the certificate, are Supplement Table S9. Of the six pre-registered predictions about the editor, one was falsified and is kept: P2, that K-draw floors would remove at least 0.4 L1 of the apparent effect, came in at +0.023, so that selection route does not explain the effect.

7 Related work

Harmonization and its protocol. iHarmony4 and DoveNet [12] define the task and the lists we audit. The checkpoints we score are DoveNet, Harmonizer [13], PCT-Net [8], DucoNet [9] and, descriptively, DiffHarmony [2]. ccHarmony [6] is the external test set of Section 4.5. R2R [11] is a recent latent-diffusion harmonizer. We did not survey which published methods evaluate on identity-disjoint lists. Ramos et al. [10] detect leakage between training data and evaluation benchmarks with image retrieval, and recommend removing leaked items from training, or failing that from the test side. We follow the test-side remedy here. In iHarmony4 the shared item is the real photograph composites are built from, which appears under different file names. We did not test retrieval-based detectors on this case.

Audits of train-test overlap. Overlap between evaluation and training data has been audited in classification, where near-duplicates between the two sides inflate measured accuracy: Barz and Denzler remove the test duplicates of CIFAR and re-score published architectures [3], and Recht et al. build fresh test sets to separate a real generalisation gap from tuning to the old one [4]. Auditing practice extends past duplicates: Northcutt et al. relabel test sets and show that rankings hold on the whole set and flip on the correctable part as the error rate rises [14], Gema et al. annotate errors in a language benchmark and release a corrected subset [15], Lewis et al. stratify question-answering test sets by overlap with training [16], Sturm documents the faults of a music-genre dataset and their effect on evaluation [17], Allamanis measures duplication between the training and evaluation splits of code models [18], and Engstrom et al. show that a gap between two populations can come from how the second population was constructed rather than from the models [19]. Kapoor and Narayanan place nonindependence between training and test samples in a taxonomy of leakage, and note that file-level checks cannot see it [5]. What none of them does is audit the construction unit of a composited benchmark, where the shared element is a photograph rather than a file.

Write-side locality. Blended Diffusion [20], DiffEdit [21], Watch Your Steps [22], ZONE [23], BrushNet [24] and EditMGT [25] localize edits and preserve the background by design, and PIE-Bench [26] and GIE-Bench [27] measure background preservation (GIE-Bench reports that GPT-Image-1 often over-modifies irrelevant regions). In the sources we read, these preservation methods bound or measure what changes rather than which outside pixels caused the change. Concurrent 2026 work addresses constrained or decomposed editing and preservation behavior: constrained edit fields and spatial edit responsibility [28], attention-scoped spatial guidance [29], decomposed efficacy and locality rewards [30], and a four-axis verifiable editing benchmark [31]. None supplies the fixed-mask, exterior-supported worst-case sensitivity certificate instantiated here. Lukoianov et al. [32] derive denoiser locality from data statistics and therefore do analyse which input neighborhoods a denoiser reads. Hu et al. [33] give a conditional, first-order leakage bound for diffusion editors under Lipschitz assumptions and label its tables illustrative. Neither supplies the fixed-mask, exterior-supported, worst-case bound on the target output instantiated here. Hu's bound is conditional rather than an instantiated constant, and both sources do address how an input region influences the output.

Certified bounds and attacks. We reuse FFT layer norms [34] and norm-preserving activations [35], developed in the Lipschitz-network literature. LipNeXt's evaluated guarantees concern classification [36], and Cert-LAS [37] uses smoothing for model-ownership verification of text-to-image diffusion, not for an editing-context certificate. In the sources we read, none certifies an out-of-region-input to in-region-output editing map. Attack-side work is owner-side and whole-image (PhotoGuard [38], immunization [39]), targets text-reading multimodal models (image-based prompt injection [40]), or triggers moderation refusals in commercial editors (MIRAGE [41]). A different method also named MIRAGE [42] performs localized multi-instance editing.

8 Limitations

What the study covers. The population contrasts rest on DoveNet, Harmonizer and our own editor, and the five-checkpoint roster adds PCT-Net and DucoNet, which follow the pooled pattern, and DiffHarmony, which reverses it. That is a study of the checkpoints we scored, not a census of the field. The added evidence is demonstration-scale: both evaluation difference-in-differences intervals span zero, with minimum detectable effects of 47.09 and 65.22 fMSE, so they are not interpretable as a null. The training-check intervals exclude zero on the training-manipulation population, and the post-tag collection of that check is a deviation rather than a change to its reading. HCOCO dominates the stratified contrast and seed uncertainty is coarse.

Departures in the intervention. The design pre-registered twelve runs with both arms of a seed pair trained in one driver process and the step budget enforced by the driver. Six of the twelve arms were interrupted and resumed. Six of the eight resume events re-applied the run seed instead of continuing the random stream, and in three arms (reduced-list DoveNet seeds 0 and 2 and full-list DoveNet seed 1) the resumed loop replayed a partial epoch so the weights received about 2.1, 2.1 and 1.0 percent more updates than the budget while the step counters still read the budget. The DoveNet trainer saves weights only, so optimizer state was not carried. Because the arms of a pair came from separate invocations, the driver's automatic equal-step check did not run. Matching seeds, step totals, row counts and weight hashes do not show that a run was uninterrupted, or that the arms differ only in training exposure. Harmonizer seeds 1 and 2 were trained on different GPUs under different torch releases, and launch commits differ across arms although the pinned scripts are byte-identical. We therefore use the saved

difference-in-differences values as descriptive results with their intervals and detection limits, and draw no causal, null or equivalence conclusion from them. Per-arm run records are in the project repository. The retrained DoveNet uses instance normalization where its leaderboard checkpoint uses batch normalization, so absolute fMSE is not comparable across the two.

Composition as an alternative explanation. The pooled matched step matches foreground ratio, not sub-dataset. Only HCOCO holds enough exposed composites for a within-sub-dataset contrast (HAdobe5k and Hday2night hold none, HFlickr holds 16 against a minimum of 30), so most of each pooled step is a contrast across sub-datasets, and inside HCOCO the MSE steps are not detected for any of the three methods, at floors of 5.9 to 15.7. That absence of detection is not a showing of no difference. Exposure inside HCOCO is not randomized, the stratified intervals are uncorrected for multiplicity, dose slopes beyond $k = 1$ did not exclude zero under the stated specifications, and the remaining post-hoc fMSE steps may reflect foreground size.

Boundaries of the audit and of the third method. The matcher is path-identity based, so near-duplicates, same-scene frames, crops and resizes carry different identity strings. The measured residue is four confirmed pairs plus one further whole-matrix candidate, five target photographs in all, 0.17 percent of the 2,917 retained backgrounds (0.14 percent for the four confirmed pairs alone), with wider-radius candidates disclosed as unadjudicated, so the retained list is not a clean split, and the cross-benchmark scan is post-hoc and name-level. The external ccHarmony comparison is mixed descriptive agreement, with that prediction unsupported at the observed point estimates and a Harmonizer-DucoNet reversal under fMSE and MSE. About 92 percent of the retained test subset had already been scored by the development run's own test-side subsets, so the full-list placement carries no fresh-test claim. The third method is a small CNN whose real-arithmetic certificate is checked by 500 Ω-confined PGD searches that never exceeded the numerical bound. The bound carries about $13\times$ slack and sits below the edit magnitude for two of five seeds. The tightness figure is an attack-dependent lower bound, and promoted weights are not bound to a pre-runtime lock. Its trained regularization penalty is not a certificate axis, being a sum of reshaped-matrix spectral norms against a product of FFT norms measured 27.7 times apart, and its FFT synthesis norm is exact only at the computed resolution, with the transfer tensor memory-bound at 5.37 GB for $n = 1024$. Shadow removal on ISTD [43] is a concluded negative whose floor-minus-typed gap was $+0.619$ $[-2.99, +4.23]$ and is not re-run here. Trained numbers from before the determinism fix carry a 1.5 L1 run-to-run noise floor (mean $|\Delta| = 1.47$) and are never aggregated with deterministic-era numbers.

9 Reproducibility and provenance

The observational analyses in this paper read saved per-composite records: the pinned re-derivation run, which stores per-composite records with foreground ratios for DoveNet and Harmonizer on all three lists and whose identifier is in Supplement S7, and the once-scored five-seed inference records of our own editor.

Provenance. The numbers in this paper are read from saved run records: per-composite score files, per-seed checkpoints bound by SHA-256, and summary files. Unless the text calls a value a registered design value (the retrain budgets of Supplement Section S12, the +479 prediction), it is a measured value. Nothing was changed in any source record or experiment for this version of the paper. No new training run was made for it, and the scorings added after the audit are inference passes over the recorded lists.

What regenerates. The cross-method contrasts, the dose statistic, the matched exposure steps and their post-hoc stratification, the near-duplicate audit and Supplement Table S10

regenerate on CPU from the re-derivation registry and per-composite records, using scripts tracked in the project repository. The table driver recomputes aggregates from per-sample rows or per-seed blocks and refuses to print a row it cannot re-derive. The two distributed lists carry recorded content hashes in Supplement S10; the staged tree, the earlier retained-only staging tree used by the gate runs, and the code fingerprint of the gate runs are recorded in the project repository. The split-identity counts of Section 3 are recomputed on CPU from the distributed lists by a short script. A second, independent pass regenerated the difference-in-differences and training-check estimates, bootstrap endpoints, standard errors and detection limits from the saved rows exactly (Supplement Section S9).

The retained list, the identity matcher and the CPU verification checks are released under the MIT license at `https://github.com/Buljic/only-means-only`. The retrain code and the baseline scoring code are not, so the retrain estimates and the population tables cannot be recomputed from the distributed artefact alone: the per-composite records they read are distributed, and the estimates are recorded in the tables.

Generative AI tools were used for standard code and language editing.

10 Conclusion

The official iHarmony4 test list has 1,399 target photographs in common with the train list, so 24.6 percent of its composites are built on a photograph the training list also uses. Pooled, that quarter scores 2 to 5 times lower in MSE for the three methods we score end to end, and the gap between two published methods is 2.3× larger on the official list in point estimates. Inside HCOCO the factors are 1.47 to 1.61, and the matched MSE steps are not detected (Harmonizer −2.86, ours −2.32, DoveNet +2.08) against pooled −62.31, −30.40 and +16.36. PCT-Net and DucoNet show the same pattern, with retained-to-excluded ratios of 4.68 and 5.73, and DiffHarmony scores worse on the shared composites in MSE while the foreground-region metric moves the other way (Supplement Table S19). We report the identity sharing, a retained test subset, a whole-matrix near-duplicate audit, and five hygiene rules. The study does not identify how much of any score difference is caused by the sharing itself. Official and retained-list scores should be reported as distinct evaluation populations.

References

[1] Wenyan Cong, Li Niu, Jianfu Zhang, Liu Liu, Zhixin Ling, Weiyuan Li, and Liqing Zhang. Image harmonization dataset iHarmony4: Generate synthetic composite images for realistic image harmonization, 2019. arXiv:1908.10526; Section 4 states the target-image separation promise.

[2] Pengfei Zhou, Fangxiang Feng, and Xiaojie Wang. DiffHarmony: Latent diffusion model meets image harmonization. In *ACM International Conference on Multimedia Retrieval (ICMR)*, 2024. arXiv:2404.06139.

[3] Björn Barz and Joachim Denzler. Do we train on test data? purging CIFAR of near-duplicates. *Journal of Imaging*, 6(6):41, 2020. arXiv:1902.00423.

[4] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In *International Conference on Machine Learning (ICML)*, 2019. arXiv:1902.10811.

[5] Sayash Kapoor and Arindam Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 2023. arXiv:2207.07048.

[6] Li Niu, Linfeng Tan, Xinhao Tao, Junyan Cao, Fengjun Guo, Teng Long, and Liqing Zhang. Deep image harmonization with globally guided feature transformation and relation distillation. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. arXiv:2308.00356; introduces the ccHarmony dataset.

[7] Yan Hong, Li Niu, Jianfu Zhang, and Liqing Zhang. Shadow generation for composite image in real-world scenes. In *AAAI Conference on Artificial Intelligence*, 2022. arXiv:2104.10338; introduces the DESOBA dataset.

[8] Julian Jorge Andrade Guerreiro, Mitsuru Nakazawa, and Björn Stenger. PCT-Net: Full resolution image harmonization using pixel-wise color transformations. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5917–5926, 2023.

[9] Linfeng Tan, Jiangtong Li, Li Niu, and Liqing Zhang. Deep image harmonization in dual color spaces. In *ACM International Conference on Multimedia (ACM MM)*, 2023. arXiv:2308.02813.

[10] Patrick Ramos, Ryan Ramos, and Noa Garcia. Data leakage in visual datasets. In *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pages 6368–6378, 2025. arXiv:2508.17416.

[11] Zhiqiu Zhang, Dongqi Fan, Mingjie Wang, Qiang Tang, Jian Yang, and Zili Yi. Region-to-region: Enhancing generative image harmonization with adaptive regional injection, 2025. arXiv:2508.09746.

[12] Wenyan Cong, Jianfu Zhang, Li Niu, Liu Liu, Zhixin Ling, Weiyuan Li, and Liqing Zhang. DoveNet: Deep image harmonization via domain verification. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. arXiv:1911.13239.

[13] Zhanghan Ke, Chunyi Sun, Lei Zhu, Ke Xu, and Rynson W. H. Lau. Harmonizer: Learning to perform white-box image and video harmonization. In *European Conference on Computer Vision (ECCV)*, 2022. arXiv:2207.01322.

[14] Curtis G. Northcutt, Anish Athalye, and Jonas Mueller. Pervasive label errors in test sets destabilize machine learning benchmarks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. arXiv:2103.14749.

[15] Aryo Pradipta Gema, Joshua Ong Jun Leang, Giwon Hong, Alessio Devoto, Alberto C. S. Mancino, Rohit Saxena, Xuanli He, Yilun Zhao, Xiaotang Yang, et al. Are we done with MMLU? In *Conference of the Nations of the Americas Chapter of the ACL (NAACL)*, 2025. arXiv:2406.04127.

[16] Patrick Lewis, Pontus Stenetorp, and Sebastian Riedel. Question and answer test-train overlap in open-domain question answering datasets. In *Conference of the European Chapter of the ACL (EACL)*, 2021. arXiv:2008.02637.

[17] Bob L. Sturm. The GTZAN dataset: Its contents, its faults, their effects on evaluation, and its future use, 2013. arXiv:1306.1461.

[18] Miltiadis Allamanis. The adverse effects of code duplication in machine learning models of code. In *ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward!)*, 2019. arXiv:1812.06469.

[19] Logan Engstrom, Andrew Ilyas, Shibani Santurkar, Dimitris Tsipras, Jacob Steinhardt, and Aleksander Madry. Identifying statistical bias in dataset replication. In *International Conference on Machine Learning (ICML)*, 2020. arXiv:2005.09619.

[20] Omri Avrahami, Dani Lischinski, and Ohad Fried. Blended diffusion for text-driven editing of natural images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. arXiv:2111.14818.

[21] Guillaume Couairon, Jakob Verbeek, Holger Schwenk, and Matthieu Cord. DiffEdit: Diffusion-based semantic image editing with mask guidance. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.11427.

[22] Ashkan Mirzaei, Tristan Aumentado-Armstrong, Marcus A. Brubaker, Jonathan Kelly, Alex Levinshtein, Konstantinos G. Derpanis, and Igor Gilitschenski. Watch your steps: Local image and scene editing by text instructions. In *European Conference on Computer Vision (ECCV)*, 2024. arXiv:2308.08947.

[23] Shanglin Li, Bohan Zeng, Yutang Feng, Sicheng Gao, Xuhui Liu, Jiaming Liu, Li Lin, Xu Tang, Yao Hu, Jianzhuang Liu, and Baochang Zhang. ZONE: Zero-shot instruction-guided local editing. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2312.16794.

[24] Xuan Ju, Xian Liu, Xintao Wang, Yuxuan Bian, Ying Shan, and Qiang Xu. BrushNet: A plug-and-play image inpainting model with decomposed dual-branch diffusion. In *European Conference on Computer Vision (ECCV)*, 2024. arXiv:2403.06976.

[25] Wei Chow, Linfeng Li, Lingdong Kong, Zefeng Li, Qi Xu, Hang Song, Tian Ye, Xian Wang, Jinbin Bai, Shilin Xu, Xiangtai Li, Junting Pan, Shaoteng Liu, Ran Zhou, Tianshu Yang, and Songhua Liu. EditMGT: Unleashing potentials of masked generative transformers in image editing. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. arXiv:2512.11715.

[26] Xuan Ju, Ailing Zeng, Yuxuan Bian, Shaoteng Liu, and Qiang Xu. PnP inversion: Boosting diffusion-based editing with 3 lines of code. In *International Conference on Learning Representations (ICLR)*, 2024. Introduces PIE-Bench; arXiv:2310.01506 (as Direct Inversion).

[27] Yusu Qian, Jiasen Lu, Tsu-Jui Fu, Xinze Wang, Chen Chen, Yinfei Yang, Wenze Hu, and Zhe Gan. GIE-Bench: Towards grounded evaluation for text-guided image editing, 2025. arXiv:2505.11493.

[28] Jingxuan Kang, Yinsong Wang, Che Liu, and Chen Qin. Constrained edit fields for training-free flow editing, 2026. arXiv:2609.33735.

[29] Zeyan Li, Wei Zhou, Hadi Amirpour, Minghao Zou, Panqi Yang, and Jianfeng Xu. Attention-scoped guidance: Training-free spatial control for image editing, 2026. arXiv:2609.37492.

[30] Shuo Chen, Fengming Huang, Yu Yao, Mingming Gong, and Tongliang Liu. Scalable minimal-change learning for controllable image editing, 2026. arXiv:2610.06021; the arXiv comment lists NeurIPS 2026.

[31] Muyao Wang, Chen Zhu, Shiqi Yang, DongHyun Hwan, and Hideki Nakayama. Beyond plausibility: Verifiable fine-grained image editing on structured assets, 2026. arXiv:2610.04381.

[32] Artem Lukoianov, Chenyang Yuan, Justin Solomon, and Vincent Sitzmann. Locality in image diffusion models emerges from data statistics, 2025. arXiv:2509.09672.

[33] Yi Hu, Leying Yi, Emily Davis, and Finn Carter. Editing on the generative manifold: A theoretical and empirical study of general diffusion-based image editing trade-offs, 2026. arXiv:2603.29736.

[34] Hanie Sedghi, Vineet Gupta, and Philip M. Long. The singular values of convolutional layers. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1805.10408.

[35] Cem Anil, James Lucas, and Roger Grosse. Sorting out Lipschitz function approximation. In *International Conference on Machine Learning (ICML)*, 2019. arXiv:1811.05381.

[36] Kai Hu, Haoqi Hu, and Matt Fredrikson. LipNeXt: Scaling up Lipschitz-based certified robustness to billion-parameter models. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2601.18513.

[37] Leyi Qi, Yiming Li, Siyuan Liang, Zhengzhong Tu, and Dacheng Tao. Cert-LAS: Toward certified model ownership verification for text-to-image diffusion models via layer-adaptive smoothing. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2605.29809.

[38] Hadi Salman, Alaa Khaddaj, Guillaume Leclerc, Andrew Ilyas, and Aleksander Madry. Raising the cost of malicious AI-powered image editing. In *International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 29894–29918, 2023.

[39] Chanhui Lee, Donggyu Choi, Seunghyun Shin, Hae-Gon Jeon, and Jeany Son. Universal image immunization against diffusion-based image editing via semantic injection, 2026. arXiv:2602.14679; the arXiv comment lists ECCV 2026.

[40] Neha Nagaraja, Lan Zhang, Zhilong Wang, Bo Zhang, and Pawan Patil. Image-based prompt injection: Hijacking multimodal LLMs through visually embedded adversarial instructions. In *International Conference on Foundation and Large Language Models (FLLM)*, pages 916–922, 2025. arXiv:2603.03637.

[41] Anshul Nasery, Ramnath Kumar, Cho-Jui Hsieh, and Sewoong Oh. MIRAGE: Protecting against malicious image editing via false moderation, 2026. arXiv:2606.26199.

[42] Ziqian Liu and Stephan Alaniz. MIRAGE: Benchmarking and aligning multi-instance image editing, 2026. arXiv:2604.05180.

[43] Jifeng Wang, Xiang Li, and Jian Yang. Stacked conditional generative adversarial networks for jointly learning shadow detection and shadow removal. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. arXiv:1712.02478; introduces the ISTD dataset.